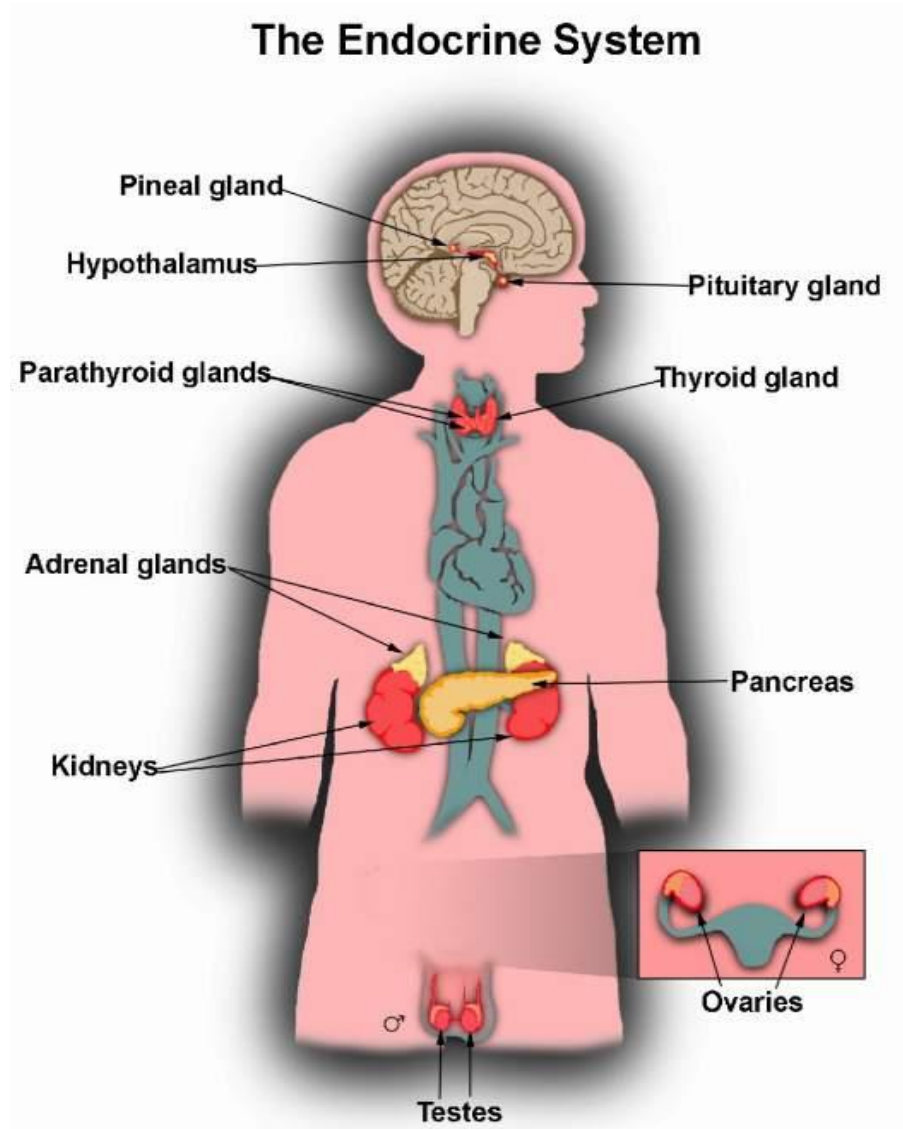


Care of Patients with Endocrine System Disorders (Diabetes Mellitus)



1. Introduction to the endocrine system:

The endocrine system consists of several glands, spreadتنتشر throughout the body. The glands release hormonesتطلق that are chemical messengers, and these hormones affect growth, development, digestion, and regulation of metabolism and reproduction. Endocrine glands generally release hormones

in the blood due to the presence of a stimuli or another hormone. The signal which the brain gives to stop the production of a particular hormone is regulated by a process called **direct feedback**, which is necessary to maintain the **homeostasis**التوازن. The body receives observations about changes in hormone levels to adjust hormone production and return to balance. When the concentration of the substance reaches the natural concentration, the gland stops producing.

The two main organs that influence the activity of the body and its effectiveness are:

1. Nervous system
2. Endocrine system

➤ **What are endocrine glands? ما هي الغدد الصماء?**

Are glandular bodies **do not have ducts** that **produce special chemicals** known as **hormones**.

Hormones: Special chemicals that secrete from the glands affect the body's various organs depending on the nature of those secretions. Some hormones are **complex proteins**, others are **simple amino acids** and **steroids**.

The glands in the body are divided into:

1. Exocrine glands :

They have ducts that discharge their secretions either inside the body such as the **salivary glands**, the **bile**, or outside the body such as: **sweat glands** and **tears**.

2. Endocrine glands :

These glands do not have channels, but they discharge directly into the blood, so their effect spreads in the body.

3. Mixed Glands:

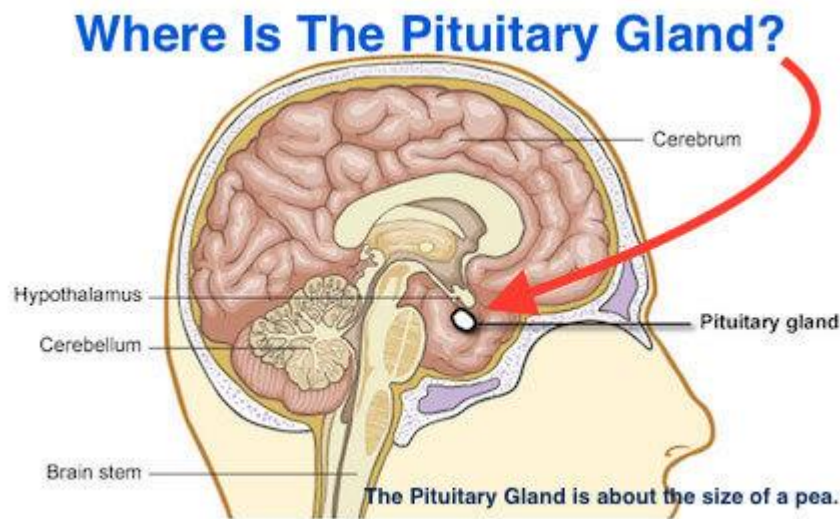
Are the glands that combine the characteristics of the two types, such as pancreatic gland and sexual glands.

- **Hormones secreted by the glands are important in performing the following functions:**
 - ✓ Organization of internal balance.
 - ✓ Body growth.
 - ✓ Sexual maturity.
 - ✓ Metabolism.
 - ✓ Human behavior and mental and intellectual development.

➤ **Endocrine in the human body:**

- 1- Pituitary gland الغدة النخامية
- 2- Thyroid gland الغدة الدرقية
- 3- Parathyroid gland الغدة جارات الدرقية
- 4- Pancreas gland غدة البنكرياس
- 5- Adrenal gland الغدة الكظرية (فوق الكلوية)
- 6- Gonads gland الغدد الجنسية
- 7- Thymus gland الغدة التيموسية (الزعترية)
- 8- Pineal gland الغدة الصنوبرية
- 9- Gastrointestinal Hormones هرمونات الجهاز الهضمي
- 10- Placental hormones هرمونات المشيمة

1. The pituitary gland الغدة النخامية:



Small-sized gland located under the brain. it is called a **master glands** because their secretions regulate the secretions of other glands.

1. The pituitary gland is divided into **three** parts:

a. **Anterior Lobe** الفص الامامي:

It forms the bulk of the pituitary gland.

❖ Hormones produced by the **anterior lobe of the pituitary gland**:

1- **Growth Hormone** :

- ✓ Increase the rate of cell division.
- ✓ Increases the transfer of amino acids in cells
- ✓ Increases the rate of protein synthesis
- ✓ Increases fat usage for energy.
- ✓ Decrease secretion of growth hormone causes **dwarfism** التقزم
- ✓ Increased secretion of the hormone causes **Gigantism** العملاقة

2- Thyroid-stimulating hormone(TSH): الهرمون المحفز للغدة الدرقية

- ✓ Increases the secretion of thyroxine and T3 increases by thyroid gland.

3. Adrenocorticotrophic hormone: هرمون القشرة الكظرية

- ✓ Increased cortisol secretion of adrenal cortex.

4. Prolactin hormone:

- ✓ It stimulates the production of milk from mammary glands.

5. Follicle-stimulating hormone:

In Women:

- ✓ Begin ovarian growth in ovarian follicles
- ✓ Increase the secretion of estrogen.

In men:

- ✓ Start producing sperm in the testicles.

6. Luteinizing hormone:

In Women:

- ✓ Causes ovulation.
- ✓ Causes ovarian follicle rupture.
- ✓ Increases the secretion of progesterone.

In men:

- ✓ The secretion of testosterone increases from the evaporation of the testicles.

7. Melanocyte-stimulating hormone (MSH): محفزة

هرمون الخلايا الصباغية

- ✓ Stimulates the production of melanin through melanocytes.

b. Intermediate pituitary gland:

- ✓ It may have something related to skin color, but his function is certainly not known.

c. Posterior Pituitary Gland:

1. Oxytocin hormone:

- ✓ Promotes uterine muscle contraction (birth).

2. Antidiuretic hormone.

- ✓ Promotes milk secretion of mammary glands.
- ✓ Increases water re-absorption through renal tubules (water returns to the blood).

2. Thyroid Gland:

a. Thyroxine and triiodothyronine hormone:

- ✓ Increase the energy production from all types of food.
- ✓ Increase protein synthesis rate.

b. Calcitonin hormone:

- Reduces absorption تقلل من امتصاص of calcium and phosphate from bone to blood.

3. Parathyroid Glands: الغدة جنب الدرقية

a. Parathyroid hormone:

- Increases the absorption of calcium and phosphate from the bones to the blood.
- Increase Calcium and phosphate absorption by small intestine.
- Increases calcium re-uptake and phosphate excretion by kidneys

4. Pancreas gland:

a. Glucagon hormone:

- Increases the conversion of glycogenic to glucose in the liver
- Increased use of excess amino acids and fat for energy.

b. Insulin hormone:

- Increases glucose transfer to cells and uses glucose to produce energy.
- Increases excess glucose conversion to the liver and muscle.
- Increases the transfer of amino acids and fatty acids to cells and use in interactions.

c. Somatostatin hormone:

- Prevents secretion of glucagon, insulin, and growth hormone

5. Adrenal Medulla gland:

a. Norepinephrine hormone:

- Causes vasoconstriction تضييق of the skin, viscera أحشاء, muscles and skeleton.

b. Epinephrine hormone:

- Increases heart rate and strength of shrinkage إنكماش
- Extends the trachea
- Reduces peristalsis (worming) in the intestine
- Increases the conversion of glycogen in the liver to glucose.
- Causes vasodilation in the muscles and skeletons.
- Causes vasoconstriction of the skin and viscera
- Increased fat use for energy.
- Increases cellular respiration rate.

6. Adrenal Cortex:

a. Aldosterone hormone:

- Re-uptake of sodium ions increases through the kidneys to the blood

b. Cortisol hormone:

- Increases the secretion of K ions through the kidneys in the urine
- Increase the use of fat and amino acids in excess of energy
- Reduces glucose use for energy (excluding brain)
- Increases glucose conversion to glycogen in the liver
- Anti-inflammatory effect: stabilizes particles and stops histamine effects

c. Androgens and estrogens hormones:

- may be involved in increased growth at puberty
- Contributing to sexual desire.
- The source of sex hormones for women after menopause

7. Ovaries:

a. Progesterone hormone:

- Increased growth and the appearance of signs of femininity (الانوثة).

8. Testicles:

a. Testosterone hormone:

- Increased growth and the appearance of masculine (الذكورة) signs

Care of Patients with Endocrine System Disorders:

1. Diabetes mellitus (DM):

Diabetes mellitus is a group of metabolic diseases characterized by **increased levels of glucose in the blood (hyperglycemia)** resulting from defects in insulin secretion, insulin action, or both.

Diabetes mellitus occurs when the pancreas doesn't make enough or any of the hormone insulin, or when the insulin produced doesn't work effectively. In diabetes, this causes the level of glucose in the blood to be too high.

Types of Diabetes mellitus:

1. Type 1 diabetes: the cells in the pancreas that make insulin are destroyed, causing a severe lack of insulin. This is thought to be the result of the body attacking and destroying its own cells in the pancreas - known as an autoimmune reaction.

It's not clear why this happens, but a number of explanations and possible triggers محفزات of this reaction have been proposed. These include:

- infection with a specific virus or bacteria.
- exposure to food-borne chemical toxins.
- exposure as a very young infant to cow's milk, where an as yet unidentified component of this triggers the autoimmune reaction in the body.

However, these are only hypotheses and are by no means proven causes.

2. Type 2 diabetes: is believed to develop when:

- The receptors on cells in the body that normally respond to the action of insulin fail to be stimulated by it - this is known as insulin resistance. In response to this more insulin may be produced, and this over-production exhausts ينهك the insulin-manufacturing cells in the pancreas;
- There is simply insufficient insulin available; and
- The insulin that is available may be abnormal and therefore doesn't work properly.

3. Gestational diabetes mellitus (GDM): is any degree of glucose intolerance with its onset during pregnancy

Onset during pregnancy, usually in the second or third trimester due to hormones secreted by the placenta المشيمة, which inhibit the action of insulin.

- ✓ Above-normal risk for perinatal complications, especially macrosomia (abnormally large babies).
- ✓ Treated with diet and, if needed, insulin to strictly maintain normal blood glucose levels.

- ✓ Occurs in about 2–5% of all pregnancies.
- ✓ Glucose intolerance **transitory** ثم تزول مؤقتة but may **recur** تعود:

- In subsequent pregnancies **في حالات الحمل اللاحقة او التالية**
 - 30–40% will develop overt diabetes (usually type 2) within 10 years (especially if obese)
- **Risk factors include:**
 - a. Obesity.
 - b. Age older than 30 years.
 - c. Family history of diabetes.
 - d. Previous large babies (>9 lb).

Screening tests (glucose challenge test) should be performed on all pregnant women between 24 and 28 wk **gestation** من الحمل

Risk Factors for Diabetes Mellitus:

- Family history of diabetes (i.e., parents or **siblings** أخوة with diabetes)
- Obesity ($\geq 20\%$ over desired body weight or BMI ≥ 27 kg/m²)
- Race/ethnicity (e.g., African Americans, Hispanic Americans, Native Americans, Asian Americans, **Pacific Islanders** جزر المحيط الهادئ)
- Age ≥ 45 y
- Previously identified impaired fasting glucose or impaired glucose tolerance
- Hypertension ($\geq 140/90$ mm Hg).
- HDL cholesterol level ≤ 35 mg/dL (0.90 mmol/L) and/or triglyceride level ≥ 250 mg/dL (2.8 mmol/L).
- History of gestational diabetes or delivery of babies over 9 lb.

Signs and Symptoms of Diabetes Mellitus:

Classic clinical manifestations of all types of diabetes include the “three Ps”:

1. **polyuria** (increased urination).
 2. **polydipsia**, (increased thirst) , occur as a result of the excess loss of fluid associated with osmotic diuresis.
 3. **polyphagia**. (increased appetite) , resulting from the catabolic state induced by insulin deficiency and the breakdown of proteins and fats.
- **Other symptoms include:**
 - a. fatigue and weakness,
 - b. sudden vision changes,

- c. tingling or numbness in hands or feet.
- d. dry skin, skin lesions or wounds that are slow to heal, and,
- e. Recurrent infections.
- f. The onset of type 1 diabetes may also be associated with sudden weight loss or nausea, vomiting, or abdominal pains, if DKA has developed.

Diagnoses:

1. The fasting blood glucose (sugar) test is the preferred way to diagnose diabetes. It is done after the person has fasted overnight (at least 8 hours).
 - Normal fasting plasma glucose levels are less than **100 milligrams** per deciliter (mg/dl).
 - Fasting plasma glucose levels of more than **126 mg/dl** on **two or more tests on different days** indicate diabetes.
 - A random blood glucose test can also be used to diagnose diabetes. A blood glucose level of 200 mg/dl or higher indicates diabetes.
2. The oral glucose tolerance test.

The procedure for oral glucose tolerance:

- The person **fasts** overnight (at least eight but not more than 16 hours).
- Then first, the fasting plasma glucose is tested.
- After this test, the person receives **75 grams of glucose**.
- Blood samples are taken at specific intervals to measure the blood glucose.
- Oral glucose tolerance test measures blood glucose levels **five times** over a period of three hours.
- In a **person without diabetes**, the glucose levels rise and then fall quickly. In someone with diabetes, glucose levels rise higher than normal and fail to come back down as fast.

Normal and Target Blood Glucose Ranges	
Normal Blood Glucose Levels in People Who Do Not Have Diabetes	
Upon waking-fasting	70 to 99 mg/dL

Normal and Target Blood Glucose Ranges	
Normal Blood Glucose Levels in People Who Do Not Have Diabetes	
After meals	70 to 140 mg/dL
Target Blood Glucose Levels in People Who Have Diabetes	
Before meals	70 to 130 mg/dL
1 to 2 hours after the start of a meal	below 180 mg/dL

Source: American Diabetes Association. Standards of Medical Care in Diabetes-2008. **Diabetes Care.** 2008;31:S12-S54.

Management of Diabetes:

Diabetes management has **five components** : مكونات

1. Nutritional therapy:

- Providing all the essential food constituents (eg, vitamins, minerals) necessary for optimal nutrition.
- Meeting energy needs.
- Achieving and maintaining a **reasonable weight** وزن معقول ..
- Preventing wide daily fluctuations** منع التقلبات اليومية الواسعة in blood glucose levels, with blood glucose levels as close to normal as is safe and practical to prevent or reduce the risk for complications.
- Decreasing serum lipid levels, if elevated, to reduce the risk for macrovascular **الاعوية الكبيرة** disease.

2. Exercise:

benefits of exercise

- Lowering blood glucose and reducing cardiovascular risk factors.

- b. Exercise lowers blood glucose levels by increasing the uptake of glucose by body muscles and by improving insulin utilization.
- c. Improves circulation and muscle tone. Resistance (strength) training, such as weight lifting, can increase **lean muscle mass** زيادة في كتلة العضلات الخالية من الدهون, thereby increasing the resting metabolic rate.

General Precautions for Exercise

- Use proper **footwear** and, if appropriate, other protective equipment.
- Avoid exercise in extreme heat or cold.
- Inspect feet daily after exercise.
- Avoid exercise during periods of poor metabolic control.

3. Monitoring, Self-Monitoring of Blood Glucose

SMBG performance include:

- Visual acuity.
- Fine motor coordination التنسيق الحركي الدقيق .
- cognitive ability,
- Comfort with technology and willingness to use it.

4. Pharmacologic therapy.

• Insulin Therapy and Insulin Preparations:

In type 1 diabetes, **exogenous insulin** انسولين خارجي must be administered for life because the body loses the ability to produce insulin.

Categories of Insulin

- A. **Short-acting insulins**, called **regular insulin** (marked R on the bottle), have an onset of 30 minutes to 1 hour; peak, 2 to 3 hours; and duration, 4 to 6 hours.
- Regular insulin is a clear solution and is usually administered 20 to 30 minutes before a meal, either alone or in combination with a longer-acting insulin.
- Regular insulin is the only insulin approved for IV use. Humulin R, Iletin Regular, and Novolin R are examples of regular insulin.

B. Intermediate-acting insulins, called NPH insulin (neutral protamine Hagedorn) or Lente insulin, have an onset of 3 to 4 hours; peak, 4 to 12 hours; and duration, 16 to 20 hours.

- Intermediate-acting insulins, which are similar in their time course of action, appear white and cloudy.
- If NPH or Lente insulin is taken alone, it is not crucial that it be taken 30 minutes before the meal.
- However, it is important that patients eat some food around the time of the onset and peak of these insulins.
- Humulin N, Iletin NPH, and Novolin N are examples of NPH insulins, and Humulin L, Iletin L, and Novolin L are examples of Lenteinsulins.

C. Long-acting insulin, such as Humulin Ultralente insulin, has a long, slow, sustained action rather than sharp, definite peaks in action. The onset of long-acting insulin is 6 to 8 hours; peak, 12 to 16 hours; and duration, 20 to 30 hours.

Oral Antidiabetic Agents

Oral antidiabetic agents may be effective for patients who have type 2 diabetes that cannot be treated effectively with MNT الحمية الغذائية and exercise alone

Examples:

- **Sulfonylureas** أدوية السلفونيل يوريا

The sulfonylureas exert their primary action by directly stimulating the pancreas to secrete insulin.

- **Thiazolidinediones** أدوية الثيازوليدينيون

Rosiglitazone (Avandia) and pioglitazone (Actos) are oral antidiabetic medications categorized as thiazolidinediones (TZDs). They are indicated for patients with type 2 diabetes who take insulin injections and whose blood glucose control is inadequate. These medications may impair liver function; therefore, liver function studies must be performed.

5. Education:

- a. Patients must learn daily self-care skills to prevent acute **fluctuations** in blood glucose.
- b. Patients must also incorporate **يديمج** into their lifestyle many preventive behaviors for avoidance of long-term diabetic complications.
- c. Patients must become **knowledgeable** **له معرفة** about nutrition, medication effects and side effects, exercise, disease progression, prevention strategies, blood glucose monitoring techniques, and medication adjustment.
- d. In addition, Patients must learn the **skills** associated with monitoring and managing diabetes and must incorporate many new activities into their daily routines.

Hypoglycemia:

Hypoglycemia (abnormally low blood glucose level) occurs when the blood glucose falls to **less than 50 to 60 mg/dL** (2.7 to 3.3 mmol/L).

Causes:

1. Too much insulin or oral hypoglycemic agents.
2. Too little food,
3. Excessive physical activity.
4. Hypoglycemia may occur at any time of the day or night. It often **occurs before meals**, especially if meals are delayed or snacks are omitted **خاصة إذا تأخرت الوجبات أو تم حذف الوجبات الخفيفة**.

Hypoglycemia is a particular concern مصدر قلق for many reasons:

- **Elderly people** frequently live alone and may not recognize the symptoms of hypoglycemia.
- With decreasing renal function, it takes longer for oral hypoglycemic agents to be excreted by the kidneys.

- Skipping mealsتخطى الوجبات may occur because of decreased appetite or financial limitations.
- Decreased visual acuityانخفاض حدة البصر may lead to errors in insulin administration.

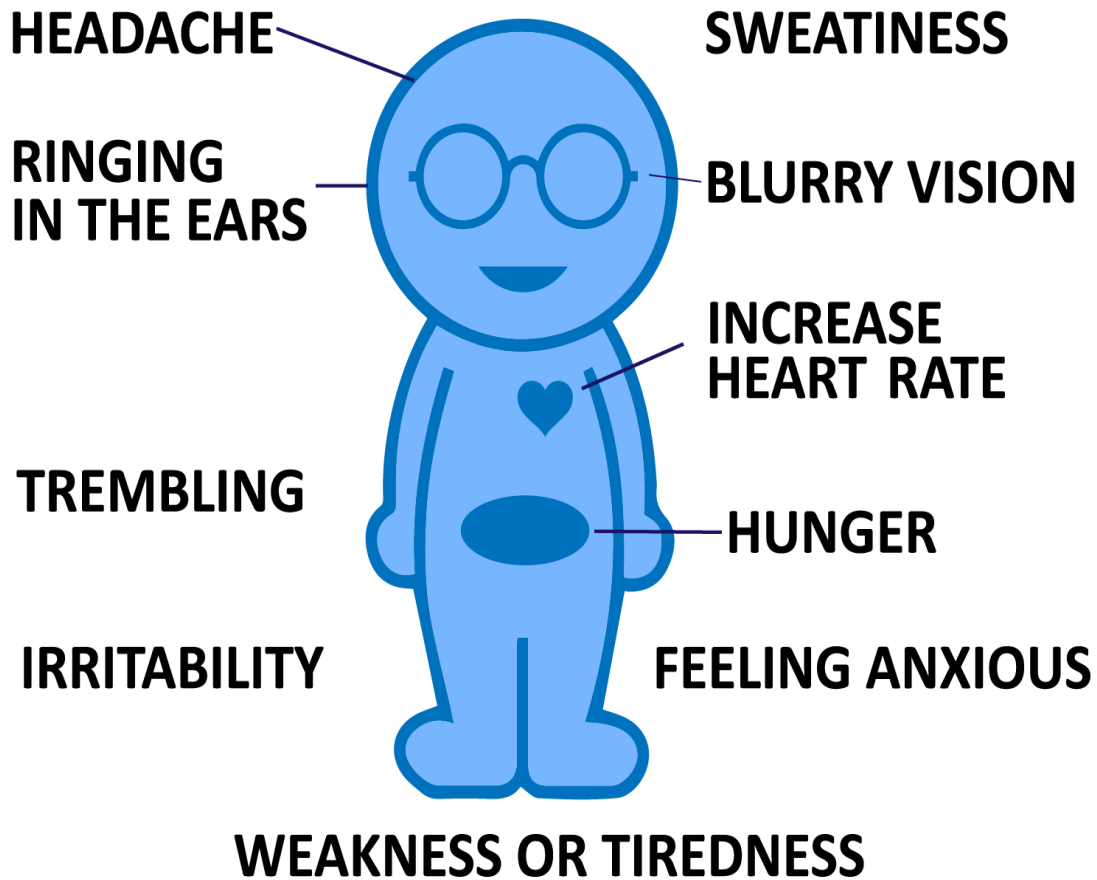
Signs and Symptoms of hypoglycemia:

In mild hypoglycemia

- Sweating.تعرق
- Tremor رجفة او رعشة
- Tachycardia.
- Palpitation.خفقان
- Nervousness, and hunger.

In moderate hypoglycemia

- Inability to concentrate.
- Headache.
- Lightheadednessدوخة, confusion مرتبك, memory lapses.
- Numbness of the lips and tongue, slurred speech تلعث في الكلام
- Impaired coordination.
- Emotional changes, irrational or combative behaviorسلوك غير عقلائي
- Double vision, and drowsiness نعاس.



In severe hypoglycemia

- Disoriented behavior. سلوك مشوش
- Seizures.
- difficulty **arousing** إيقاظ from sleep, or
- Loss of consciousness.

Complications:

1. Seizure نوبات صرع
2. Loss of consciousness فقدان الوعي
3. Death, primarily in those with diabetes

Management of hypoglycemia:

Immediate treatment must be given when hypoglycemia occurs. The usual recommendation is for **15 g of a fast-acting concentrated source of carbohydrate** such as the following, given orally:

- Three or four commercially prepared glucose tablets
- 4 to 6 oz of fruit juice or regular soda

- 6 to 10 hard candies حلوى صلبة
- 2 to 3 teaspoons of sugar or honey

Note = 1 ounce = 28.3 gm

Nursing Process for patients With Diabetes Mellitus:

A: Assessment

1. The history and physical assessment focus on the **signs and symptoms** of prolonged **hyperglycemia** and on physical, social, and emotional factors that may affect the patient's ability to learn and perform diabetes self-care activities.
2. Assessed for physical factors that may impair his or her ability to learn or perform self-care skills, such as, **Visual deficits**, **Deficits in motor coordination**, and typical daily schedule.

B: Diagnosis

- Risk for **fluid volume deficit** related to polyuria and dehydration
- **Imbalanced nutrition** related to imbalance of insulin, food, and physical activity
- **Deficient knowledge** about diabetes self-care skills/information
- Potential self-care deficit related to physical impairments or social factors
- **Anxiety** **القلق** related to loss of control, fear of inability to manage diabetes, misinformation related to diabetes, fear of diabetes complications.

C: Planning and Goals

1. Maintenance of fluid and electrolyte balance.
2. Optimal control of blood glucose levels.
3. Reversal of weight loss.
4. ability to perform diabetes survival skills and self-care activities, decreased anxiety, and,
5. Absence of complications.

D: Nursing Interventions:

1. Maintaining Fluid and Electrolyte Balance

- Intake and output are measured.
- IV fluids and electrolytes are administered as prescribed, and
- Oral fluid intake is encouraged when it is permitted.

2. Improving Nutritional Intake:

1. An appropriate **caloric intake** allows the patient to achieve and maintain the desired body weight.
 - a. The patient is encouraged to **eat full meals and snacks** as prescribed in the diet prescription.
 - b. Arrangements may be made with the **dietitian** for extra snacks before increased physical activity.
2. Emotional support to reduce anxiety.
3. Monitoring and Managing Potential Complications
 - زيادة السوائل Fluid Overload
 - نقص بوتاسيوم الدم Hypokalemia
 - Hypoglycemia

E: Evaluation

1. Achieves fluid and electrolyte balance:

- a. Demonstrates intake and output balance
- b. Exhibits electrolyte values within normal limits
- c. Exhibits vital signs that remain stable, with resolution of **orthostatic hypotension** and tachycardia

2. Achieves metabolic balance:

- a. Avoids **extremes** زيادة او نقصان of glucose levels (hypoglycemia or hyperglycemia)
- b. Exhibits glucose levels within target range with minimal episodes of hypoglycemia

3. Demonstrates/verbalizes diabetes survival skills.

4. Absence of complications:

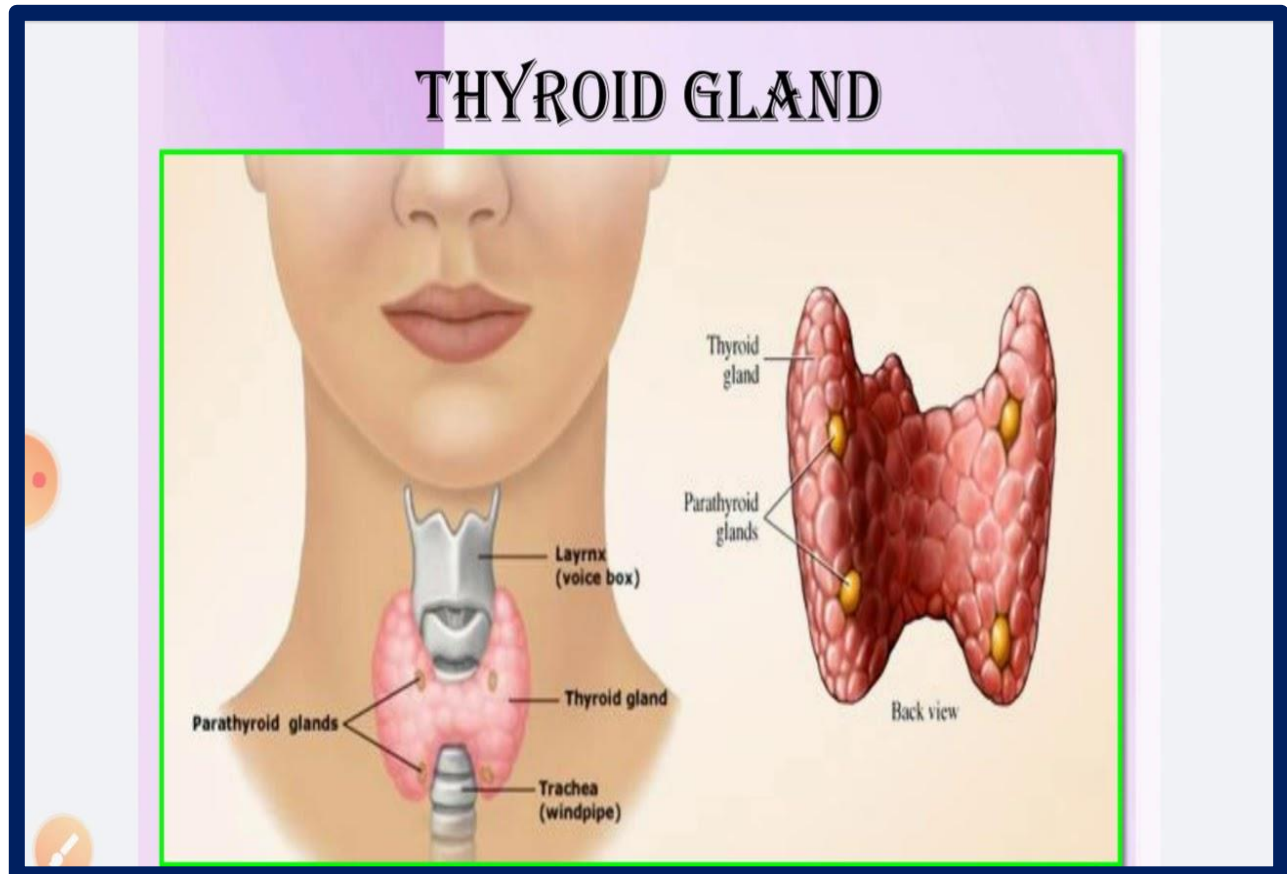
- a. Exhibits **normal cardiac rate and rhythm and normal breath sounds**
- b. Exhibits no jugular venous distention عدم وجود انتفاخ في الوريد الوداجي في الرقبة
- c. Exhibits blood glucose and urine ketone levels within target range.
- d. Exhibits no manifestations of hypoglycemia or hyperglycemia.
- e. Shows improved mental status without signs of cerebral edema.

jugular venous distention



Hyperthyroidism:

- Hyperthyroidism is a condition in which an overactive thyroid gland is producing **an excessive amount of thyroid hormones** that circulate in the blood.
- Graves' disease, the most common type of hyperthyroidism, results from an excessive output of thyroid hormones caused by abnormal stimulation of the thyroid gland by circulating immunoglobulins.
- Thyrotoxicosis is a toxic condition that is caused by an excess of thyroid hormones from any cause.



Causes of hyperthyroidism:

Some common causes of hyperthyroidism include:

1. Graves' Disease.
2. Functioning adenoma ("hot nodule") and toxic multi-nodular goiter (TMNG) .
3. Excessive intake of thyroid hormones.
4. Abnormal secretion of TSH.
5. Thyroiditis (inflammation of the thyroid gland) .
6. Excessive iodine intake.

Graves' Disease:

Graves' disease, a condition named for an Irish doctor who first described the condition. This condition can be summarized by noting that an enlarged thyroid (enlarged thyroids are called **goiters**) is producing way too much thyroid hormone. (Remember that only a small percentage of goiters produce too much thyroid hormone; the majority of thyroid goiters actually become large because they are not producing enough thyroid hormone.)

There are actually three distinct parts of Graves' disease:

1. Over activity of the thyroid gland (hyperthyroidism)

2. Inflammation of the tissues around the eyes, causing swelling.
3. Thickening of the skin over the lower legs (pretibial myxedema).

Clinical manifestations of hyperthyroidism(S&S):

System	S&S
Cardiovascular	Tachycardia, palpitations, increased cardiac output, Warm skin, heat intolerance
Neurological	Fatigue, restlessness, tremor, insomnia, emotional Instability, muscle weakness, staring gaze. التحديق في النظر
Pulmonary	Dyspnea
Integumentary (Skin)	Diaphoresis; warm, moist skin; fine, soft hair, hair loss
Gastrointestinal	Increased appetite, weight loss, frequent stools, decreased serum lipid levels, Increased bowel movements
Reproductive	Decreased libido, erectile dysfunction, amenorrhea

Diagnosis of hyperthyroidism:

Hyperthyroidism can be suspected in patients with:

- tremors,
- excessive sweating,
- smooth velvety skin,
- fine hair,
- a rapid heart rate, and
- an enlarged thyroid gland.
- the diagnosis is made on the basis of the symptoms,
 - a decrease in serum TSH.
 - increased free T₄, and ,
 - an increase in radioactive iodine uptake.



Figure shows thyroid screening (palpation)

Treatment:

Pharmacologic Therapy

Two forms of pharmacotherapy are available for treating hyperthyroidism and controlling excessive thyroid activity:

1. Use of irradiation by administration of the radioisotope iodine for destructive effects on the thyroid gland and,
2. Antithyroid medications that interfere with the synthesis of thyroid hormones and other agents that control manifestations of hyperthyroidism.
 - Propylthiouracil (PTU), Blocks synthesis of hormones (conversion of T_3 to T_4).
 - Methimazole, Blocks synthesis of thyroid hormone.
 - Dexamethasone, Suppresses release of thyroid hormone.
 - propranolol and digitalis may be prescribed to treat cardiac symptoms.

Note

Patients receiving these medications should be observed for the development of goiter and should be cautioned against use of over-the-counter medications that contain iodides and can increase the response to iodide therapy. Cough medications, expectorants, bronchodilators, and salt substitutes may contain iodide and should be avoided by the patient receiving iodide therapy.

A. Surgical treatment

- in patients with large goiters, or in patients who are unable to take antithyroid agents. Surgery for treatment of hyperthyroidism is performed soon after the thyroid function has returned to normal (4 to 6 weeks).
- The surgical removal of about five sixths of the thyroid tissue (subtotal thyroidectomy) reliably results in a prolonged remission in most patients with exophthalmic goiter.

Nursing Process for Patient with Hyperthyroidism:

A: Assessment

1. Assess the patient's nutritional status and the presence of symptoms. Symptoms related to excessive nervous system output and changes in vision and appearance of the eyes are noted.
2. Assesses and monitors the patient's cardiac status, including heart rate, blood pressure, heart sounds, and peripheral pulses.
3. Assess the emotional state and psychological status.

B: Diagnosis

- Imbalanced nutrition, less than body requirements, related to exaggerated metabolic rate, excessive appetite, and increased gastrointestinal activity
- Ineffective coping related to irritability, hyperexcitability فرط الاستثارة, apprehension خوف, and emotional instability
- Low self-esteem عدم الاهتمام بنفسه related to changes in appearance, excessive appetite, and weight loss
- Altered body temperature

C: Planning and Goals

1. Improved nutritional status.
2. Improved coping ability.
3. Improved self-esteem.
4. Maintenance of normal body temperature, and ,
5. Absence of complications.

D: Nursing Interventions:

1. Improving Nutritional Status

- a. Advice patient and family to several well-balanced meals of small size, even up to six meals a day.
- b. To reduce diarrhea, highly seasoned foods and stimulants such as coffee, tea, cola, and alcohol are discouraged.
- c. High-calorie, high-protein foods are encouraged.

2. Enhancing Coping Measures

- a. Use a calm, unhurried approach with the patient.
- b. Stressful experiences are minimized
- c. The environment is kept quiet and uncluttered. Noises, such as loud music, conversation, and equipment alarms, are minimized.

3. Improving Self-Esteem

4. Adjust the room temperature to maintaining normal body temperature.

5. Monitoring and Managing Potential Complications.

- a. Cardiac and respiratory functions are assessed by measuring vital signs and cardiac output, electrocardiographic (ECG) monitoring, arterial blood gases, and pulse oximetry.
- b. Oxygen is administered to prevent hypoxia, to improve tissue oxygenation, and to meet the high metabolic demands.
- c. IV fluids may be necessary to maintain blood glucose levels and to replace lost fluids.

E: Evaluation:

1. Improves nutritional status

- a. Reports adequate dietary intake and decreased hunger.
- b. Avoids use of alcohol and other stimulants.
- c. Reports decreased episodes of diarrhea.

2. Demonstrates effective coping methods in dealing with family, friends, and coworkers.

- a. Explains reasons for irritability and emotional instability
- b. Avoids stressful situations, events, and people
- c. Participates in relaxing, non-stressful activities.

3. Achieves increased self-esteem

- a. Verbalizes feelings about self and illness
- b. Describes feelings of frustration and loss of control
- c. Describes reasons for increased appetite

4. Maintains normal body temperature

5. Absence of complications:

- a. Has serum thyroid hormone and TSH levels within normal limits
- b. Identifies signs and symptoms of thyroid storm and hypothyroidism
- c. Has vital signs and results of ECG, arterial blood gases, and pulse oximetry within normal limits
- d. States importance of regular follow-up and lifelong maintenance of prescribed therapy

Hypothyroidism:

- **Hypothyroidism is a condition characterized by abnormally low thyroid hormone production.**
- There are many disorders that result in hypothyroidism.
- These disorders may directly or indirectly involve the thyroid gland.
- Because thyroid hormone affects growth, development, and many cellular processes, inadequate thyroid hormone has widespread consequences for the body.
- Hypothyroidism results from suboptimal levels of thyroid hormone.
- Thyroid deficiency can affect all body functions and can range from mild, subclinical forms to myxedema, an advanced form

Causes of Hypothyroidism:

1. Autoimmune disease (Hashimoto's thyroiditis, post-Graves' disease)

2. Atrophy of thyroid gland with aging
3. Therapy for hyperthyroidism
 - Radioactive iodine.
 - Thyroidectomy
4. Medications
 - Lithium.
 - Iodine compounds.
 - Antithyroid medications
5. Radiation to head and neck for treatment of head and neck cancers, lymphoma
6. Infiltrative diseases of the thyroid (amyloidosis, scleroderma, lymphoma)
7. Iodine deficiency and iodine excess.

Clinical manifestations of hypothyroidism(S&S):

System	S&S
Cardiovascular	Bradycardia, decreased cardiac output, cool skin, cold intolerance
Neurological	Lethargyخمول, slowed movements, memory loss, confusion
Pulmonary	Dyspnea, hypoventilation
Integumentary (Skin)	Cool, dry skin; brittle, dry hair
Gastrointestinal	Decreased appetite, weight gain, constipation, increased serum lipid levels
Reproductive	Decreased libido, erectile dysfunction

Medical Management:

Synthetic levothyroxine (Synthroid or Levothroid) is the preferred preparation for treating hypothyroidism and suppressing nontoxic goiters.

Note:

The nurse must monitor for myocardial ischemia or infarction, which can occur in response to therapy in patients with severe, long-standing hypothyroidism or myxedema coma. The nurse must also be alert for signs of angina, especially during the early phase of treatment; if detected, it must be reported and treated at once to avoid a fatal myocardial infarction.

Plan of Nursing Care: Care of the Patient with Hypothyroidism:

A: Assessment

- Health history
- Physical examination

B: Diagnosis

- Activity intolerance related to fatigue and depressed cognitive process
- Risk for imbalanced body temperature.
- Constipation related to depressed gastrointestinal function.
- Deficient knowledge about the therapeutic regimen for lifelong thyroid replacement therapy.
- Ineffective breathing pattern related to depressed ventilation.
- Disturbed thought processes related to depressed metabolism and altered cardiovascular and respiratory status.

C: Planning and goals

1. Increased participation المشاركة in activities and increased independence.
2. Maintenance of normal body temperature.
3. Return of normal bowel function.
4. Knowledge and acceptance of the prescribed therapeutic regimen.
5. Improved respiratory status and maintenance of normal breathing pattern.
6. Improved thought processes.

D: Implementation

1. Promote independence in self-care activities. Space activities to promote rest and exercise as tolerated.
2. Monitor patient's response to increasing activities.
3. Provide extra layer of clothing or extra blanket.
4. Avoid and discourage use of external heat source (eg, heating pads, electric or warming blankets).
5. Protect from exposure to cold and drafts.
6. Encourage increased fluid intake within limits of fluid restriction, Provide foods high in fiber, foods with high water content.
7. Explain rationale for thyroid hormone replacement.
8. Explain the necessity for long-term follow-up to patient and family.

9. Monitor respiratory rate, depth, pattern, pulse oximetry, and arterial blood gases.
10. Encourage deep breathing, coughing, and use of incentive spirometry.
11. Orient patient to time, place, date, and events around him or her.

Hyperparathyroidism:

Hyperparathyroidism, caused by **overproduction** of parathyroid hormone (PTH) by the parathyroid glands with a subsequent increase in the serum calcium level (hypercalcemia), is characterized by bone decalcification انحلال العظام and the development of renal calculi (kidney stones) containing calcium.

- **Primary hyperparathyroidism** occurs two to four times more often in **women** than in men and is most common in people between 60 and 70 years of age. The disorder is rare in children younger than 15 years of age, but the incidence increases tenfold between the ages of 15 and 65 years. Half of the people diagnosed with hyperparathyroidism do not have symptoms.
- **Secondary hyperparathyroidism**, with manifestations similar to those of primary hyperparathyroidism, occurs in patients who have chronic renal failure and so-called renal rickets as a result of phosphorus retention, increased stimulation of the parathyroid glands, and increased parathyroid hormone secretion.

Clinical Manifestations:

1. The patient may have no symptoms.
2. Apathy خمول.
3. Fatigue, muscle weakness.
4. Nausea, anorexia, vomiting, constipation.
5. Hypertension, and cardiac dysrhythmias may occur.
6. Irritability and neurosis to psychoses caused by the direct effect of calcium on the brain and nervous system.
7. Because calcium is being removed from bones, joint pain and pathological fractures may occur. With severe hypercalcemia.
8. The increased serum calcium level also causes gastrin secretion, resulting in abdominal pain and peptic ulcers.

Diagnostic Findings

1. Persistent elevation of serum calcium levels and an elevated concentration of parathyroid hormone.
2. X-ray or bone scans.
3. The double-antibody parathyroid hormone test is used to distinguish between primary hyperparathyroidism and malignancy as a cause of hypercalcemia.
4. Ultrasound, MRI, thallium scan, and fine-needle biopsy have been used to evaluate the function of the parathyroids and to localize parathyroid cysts, adenomas, or hyperplasia.

Medical Management, which include:

1. Hydration Therapy:

- a. Patients with hyperparathyroidism are at risk for renal calculi. Therefore, a daily fluid intake of 2000 mL or more is encouraged to help prevent calculus formation.
 - b. Cranberry juice is suggested, because it may lower the urinary pH.
 - c. Thiazide diuretics are avoided, because they decrease the renal excretion of calcium and further elevate serum calcium levels.
 - d. Avoid dehydration.
2. Walking or use of a rocking chair for those with limited mobility, is encouraged as much as possible, because bones that are subjected to normal stress give up less calcium.
 3. the patient is advised to avoid a diet with restricted or excess calcium.
 4. Improve the appetite and prevent constipation.

Hypoparathyroidism:

The most common cause of hypoparathyroidism is **inadequate secretion** of parathyroid hormone (PTH) after interruption of the blood supply or surgical removal of parathyroid gland tissue during thyroidectomy, parathyroidectomy, or radical neck dissection.

Clinical Manifestations:

- 1- Tetany is a general muscle hypertonia.(numbness, tingling, and cramps in the extremities, and the patient complains of stiffness in the hands and feet).
- 2- Anxiety, irritability, depression, and even delirium.
- 3- ECG changes and hypotension also may occur.

Diagnostic Findings

1.Trousseau's sign is positive when carpopedal spasm يتشنجات في رسخ اليد is induced by occluding انسداد the blood flow to the arm for 3 minutes with a blood pressure cuff.

2.Chvostek's sign is positive when a sharp tapping over the facial nerve just in front of the parotid gland and anterior to the ear causes spasm or twitching of the mouth, nose, and eye.

3. X-rays of bone show increased density.

4. Calcification is detected on x-rays .

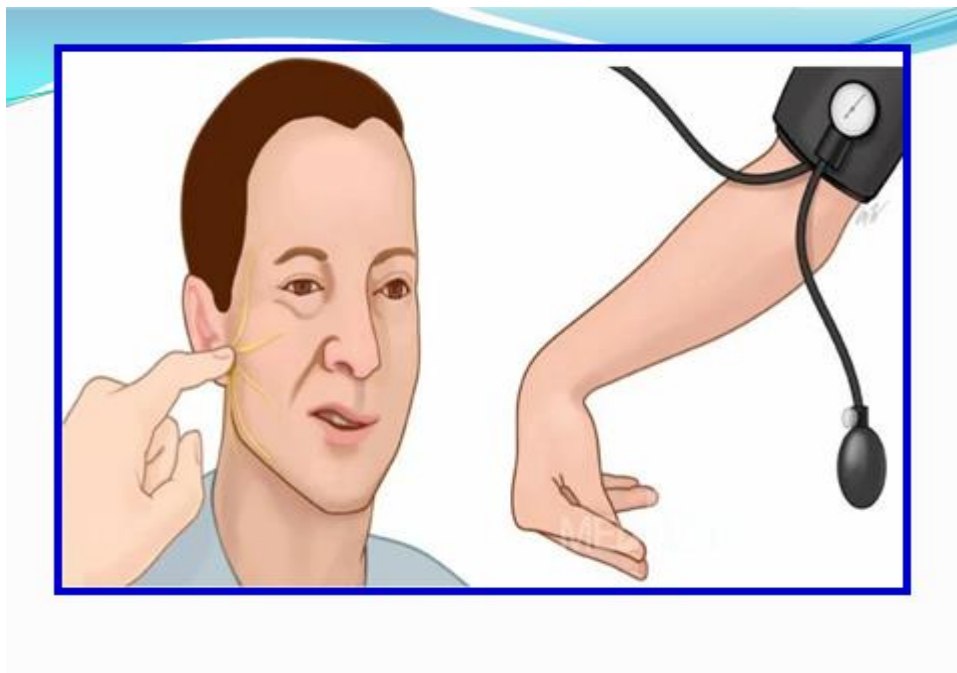


Figure: Trousseau's sign and Chvostek's sign

Medical Management

The goal of therapy is to increase the serum calcium level to 9 to 10 mg/dL (2.2 to 2.5 mmol/L) and to eliminate the symptoms of hypoparathyroidism and hypocalcemia. When hypocalcemia and tetany occur after a thyroidectomy:

1. The immediate treatment is administration of IV calcium gluconate.
2. If this does not decrease neuromuscular irritability and seizure activity immediately, sedative agents such as pentobarbital may be administered.
3. Parenteral parathormone can be administered to treat acute hypoparathyroidism.
4. A diet high in calcium and low in phosphorus is prescribed.
5. Although milk, milk products, and egg yolk are high in calcium, they are restricted because they also contain high levels of phosphorus.
6. Variable dosages of a vitamin D preparation are usually required and enhance calcium absorption from the gastrointestinal tract.

حكمة:

أظهر الناس أعراقاً أحسنهم أخلاقاً